

Day 16: Decision Tree — Complete ID3 Calculation

Step-by-Step Walkthrough with the Tennis Dataset

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The Tennis Dataset (14 Examples)

#	Outlook	Temp	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Target: PlayTennis — Yes (9), No (5)

Step 1: What is Entropy?

Entropy measures **how mixed** (impure) a set is.

$$H(S) = - \sum_{i=1}^c p_i \log_2(p_i)$$

Intuition:

- If all examples are Yes $\rightarrow H = 0$ (no confusion, pure)
- If half Yes, half No $\rightarrow H = 1$ (maximum confusion)
- More mixed = higher entropy = harder to predict

Think of it as

“How surprised would I be if I randomly picked one example?”

Step 2: Calculate Entropy of the Entire Dataset

Our dataset has 14 examples: **9 Yes, 5 No**

$$p(\text{Yes}) = \frac{9}{14} = 0.643, \quad p(\text{No}) = \frac{5}{14} = 0.357$$

$$H(S) = -p(\text{Yes}) \log_2 p(\text{Yes}) - p(\text{No}) \log_2 p(\text{No})$$

$$= -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14}$$

$$= -(0.643)(-0.637) - (0.357)(-1.485)$$

$$= 0.410 + 0.530 = \boxed{0.940}$$

Step 3: Information Gain — The Key Idea

Information Gain = How much does entropy **decrease** if we split on a feature?

$$IG(S, A) = H(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} \cdot H(S_v)$$

In plain English:

- 1 Start with total entropy (0.940)
- 2 Split the data by one feature (e.g., Outlook)
- 3 Calculate entropy of **each group** after the split
- 4 Compute the **weighted average** of those group entropies
- 5 **Subtract** from total entropy

Higher IG = better split = that feature tells us more!

Step 4a: Information Gain for Outlook

Split the 14 examples by Outlook:

Outlook	Yes	No	Total
Sunny	2	3	5
Overcast	4	0	4
Rain	3	2	5
Total	9	5	14

Now calculate entropy for each group:

- $H(\text{Sunny}) = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.971$
- $H(\text{Overcast}) = -\frac{4}{4} \log_2 \frac{4}{4} = 0$ (pure! all Yes)
- $H(\text{Rain}) = -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} = 0.971$

Step 4a (continued): IG(Outlook) Calculation

Weighted average entropy after splitting on Outlook:

$$\begin{aligned}H(S|\text{Outlook}) &= \frac{5}{14} \cdot H(\text{Sunny}) + \frac{4}{14} \cdot H(\text{Overcast}) + \frac{5}{14} \cdot H(\text{Rain}) \\&= \frac{5}{14} \times 0.971 + \frac{4}{14} \times 0 + \frac{5}{14} \times 0.971 \\&= 0.347 + 0 + 0.347 \\&= 0.694\end{aligned}$$

Information Gain:

$$IG(\text{Outlook}) = H(S) - H(S|\text{Outlook}) = 0.940 - 0.694 = \boxed{0.247}$$

Step 4b: Information Gain for Temperature

Split by Temperature:

Temperature	Yes	No	Total
Hot	2	2	4
Mild	4	2	6
Cool	3	1	4

Entropy of each group:

- $H(\text{Hot}) = -\frac{2}{4} \log_2 \frac{2}{4} - \frac{2}{4} \log_2 \frac{2}{4} = 1.0$
- $H(\text{Mild}) = -\frac{4}{6} \log_2 \frac{4}{6} - \frac{2}{6} \log_2 \frac{2}{6} = 0.918$
- $H(\text{Cool}) = -\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4} = 0.811$

$$H(S|\text{Temp}) = \frac{4}{14}(1.0) + \frac{6}{14}(0.918) + \frac{4}{14}(0.811) = 0.911$$

$$IG(\text{Temp}) = 0.940 - 0.911 = \boxed{0.029}$$

Step 4c: Information Gain for Humidity

Split by Humidity:

Humidity	Yes	No	Total
High	3	4	7
Normal	6	1	7

Entropy of each group:

- $H(\text{High}) = -\frac{3}{7} \log_2 \frac{3}{7} - \frac{4}{7} \log_2 \frac{4}{7} = 0.985$
- $H(\text{Normal}) = -\frac{6}{7} \log_2 \frac{6}{7} - \frac{1}{7} \log_2 \frac{1}{7} = 0.592$

$$H(S|\text{Humidity}) = \frac{7}{14}(0.985) + \frac{7}{14}(0.592) = 0.788$$

$$IG(\text{Humidity}) = 0.940 - 0.788 = \boxed{0.152}$$

Step 4d: Information Gain for Wind

Split by Wind:

Wind	Yes	No	Total
Weak	6	2	8
Strong	3	3	6

Entropy of each group:

- $H(\text{Weak}) = -\frac{6}{8} \log_2 \frac{6}{8} - \frac{2}{8} \log_2 \frac{2}{8} = 0.811$
- $H(\text{Strong}) = -\frac{3}{6} \log_2 \frac{3}{6} - \frac{3}{6} \log_2 \frac{3}{6} = 1.0$

$$H(S|\text{Wind}) = \frac{8}{14}(0.811) + \frac{6}{14}(1.0) = 0.892$$

$$IG(\text{Wind}) = 0.940 - 0.892 = \boxed{0.048}$$

Step 5: Compare All Information Gains — Pick the Root

Feature	Information Gain
Outlook	0.247 ← HIGHEST!
Humidity	0.152
Wind	0.048
Temperature	0.029

Decision

Outlook has the highest Information Gain, so it becomes the **root node** of our decision tree!

We now split the dataset into 3 branches: Sunny (5), Overcast (4), Rain (5)

Step 6: The Overcast Branch — Done!

Examples where Outlook = **Overcast**: rows 3, 7, 12, 13

#	Temp	Humidity	Wind	Play?
3	Hot	High	Weak	Yes
7	Cool	Normal	Strong	Yes
12	Mild	High	Strong	Yes
13	Hot	Normal	Weak	Yes

All 4 examples are Yes!

$H(\text{Overcast}) = 0$ (perfectly pure)

Result

Overcast → **Yes** (leaf node — no more splitting needed!)

Step 7: The Sunny Branch — Need More Splitting

Examples where Outlook = **Sunny**: rows 1, 2, 8, 9, 11

#	Temp	Humidity	Wind	Play?
1	Hot	High	Weak	No
2	Hot	High	Strong	No
8	Mild	High	Weak	No
9	Cool	Normal	Weak	Yes
11	Mild	Normal	Strong	Yes

This subset has 2 Yes, 3 No $\rightarrow H(\text{Sunny}) = 0.971$ (not pure!)

We need to find the best feature to split this subset. Remaining features: **Temperature, Humidity, Wind**

Step 7 (continued): IG Within the Sunny Branch

For the 5 Sunny examples (2 Yes, 3 No), calculate IG for each remaining feature:

Humidity:

- High: 0 Yes, 3 No $\rightarrow H = 0$ (pure!)
- Normal: 2 Yes, 0 No $\rightarrow H = 0$ (pure!)

$$H(S_{\text{Sunny}}|\text{Humidity}) = \frac{3}{5}(0) + \frac{2}{5}(0) = 0, \quad IG(\text{Humidity}) = 0.971 - 0 = \mathbf{0.971}$$

Temperature: $IG = 0.571$ **Wind:** $IG = 0.020$

Decision

Humidity has the highest IG (0.971), so we split on Humidity!

High $\rightarrow$, Normal $\rightarrow$

Step 8: The Rain Branch — Need More Splitting

Examples where Outlook = **Rain**: rows 4, 5, 6, 10, 14

#	Temp	Humidity	Wind	Play?
4	Mild	High	Weak	Yes
5	Cool	Normal	Weak	Yes
6	Cool	Normal	Strong	No
10	Mild	Normal	Weak	Yes
14	Mild	High	Strong	No

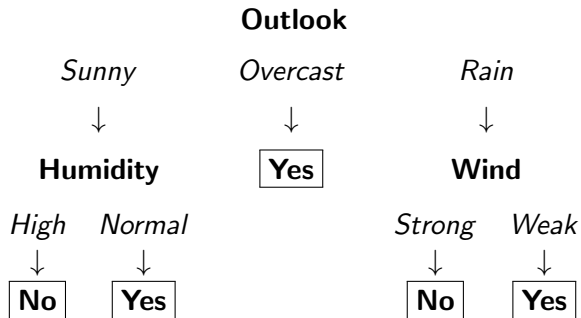
This subset has 3 Yes, 2 No $\rightarrow H(\text{Rain}) = 0.971$

Wind: Weak (3 Yes, 0 No) $\rightarrow H = 0$; Strong (0 Yes, 2 No) $\rightarrow H = 0$
 $IG(\text{Wind}) = 0.971 - 0 = \mathbf{0.971}$ (perfect split!)

Decision

Split on **Wind**: Weak $\rightarrow$, Strong $\rightarrow$

The Complete Decision Tree



This tree correctly classifies all 14 training examples with **100% accuracy**.

How to Read the Tree — Making Predictions

Example 1: Sunny, Cool, Normal humidity, Strong wind. Play tennis?

- 1 Start at root: **Outlook?** → Sunny
- 2 Go to Sunny branch: **Humidity?** → Normal
- 3 Reach leaf: **Yes** — Play tennis!

Example 2: Rain, Mild, High humidity, Strong wind. Play tennis?

- 1 **Outlook?** → Rain
- 2 **Wind?** → Strong
- 3 Reach leaf: **No** — Don't play!

Notice

Temperature was **never used**! It had the lowest IG, so the tree found better features.

The ID3 Algorithm — Summary

- ① **Input:** Dataset with features and a target class
- ② **Calculate** entropy of the full dataset
- ③ **For each feature**, calculate Information Gain
- ④ **Pick** the feature with the highest IG → make it the current node
- ⑤ **Split** the dataset into subsets (one per feature value)
- ⑥ **For each subset:**
 - If all examples have the same class → create a **leaf** with that class
 - Otherwise → **recurse** (go back to step 3 with the subset)

Key Properties of ID3

- **Greedy:** always picks the locally best feature (no backtracking)
- **Top-down:** builds tree from root to leaves
- **Recursive:** same process repeats at every node

Why Decision Trees Are Powerful

Strengths

- Easy to explain to anyone
- No feature scaling needed
- Handles categorical data
- Shows feature importance
- Fast predictions

Weaknesses

- Can overfit (memorize noise)
- Unstable with small changes
- Greedy $\neq$ globally optimal
- Cannot learn diagonal boundaries

Solution to Overfitting

Random Forest = build many trees on random subsets and let them vote!

Quick Recap — What We Learned

- ① **Entropy** = measure of impurity ($H = -\sum p_i \log_2 p_i$)
- ② **Information Gain** = reduction in entropy after a split
- ③ **ID3 Algorithm**: always split on the feature with highest IG
- ④ For the Tennis dataset:
 - Root = **Outlook** (IG = 0.247)
 - Sunny branch splits on **Humidity** (IG = 0.971)
 - Rain branch splits on **Wind** (IG = 0.971)
 - Overcast = pure **Yes** (no split needed)
- ⑤ The final tree has **depth 2** and classifies all 14 examples correctly